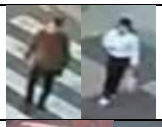
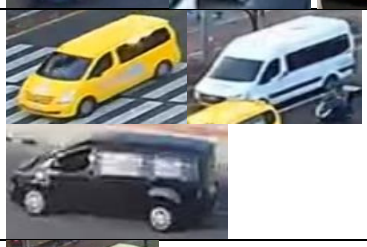
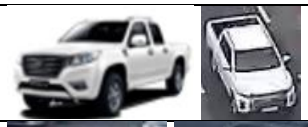
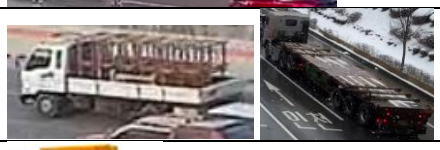
TSBOW

Traffic Surveillance Benchmark for Occluded vehicles under various Weather conditions

CLASS DEFINITION

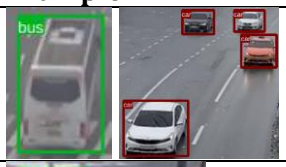
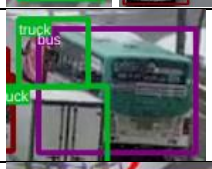
Class Definition

cat_id	category	train_id	id	name	description	example
0	void	255	0	unidentified	unclear or corrupted	
0	void	255	1	others	Any vehicles that are clearly visible but not included in this list	
1	pedestrian	0	2	pedestrian		
2	micromobility	1	3	micromobility	cycles (bike, motorbike)	
					scooter	
3	vehicle	2	4	car	sedan (2-4 seats)	
					SUV (6 seats)	
					van (7-10 seats)	
3	vehicle	3	5	bus	mini bus (30 seats)	
					bus (50 seats)	

3	vehicle	4	6	small truck (same size as an SUV or van)	pickup	
					small truck	
					small box truck	
3	vehicle	5	7	truck	large box truck	
					trailer truck	
					flatbed truck	
					dump truck	
					tanker truck	
					concrete truck	
					garbage truck	
					crane truck	
					tow truck	

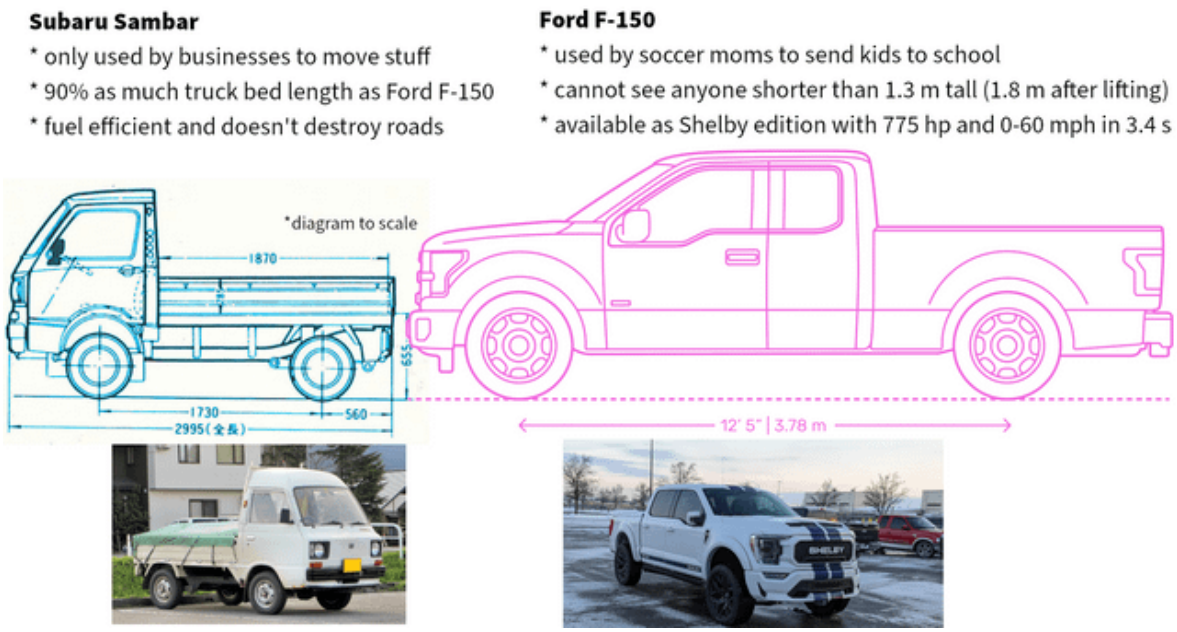
					fire truck	
					car transporter	

Labelling Policy

#	Policy	Example
1	**Draw bounding boxes tightly around the objects of interest.** This helps the model accurately learn the object's shape and location. Avoid unnecessary background or empty space within the bounding box.	
2	**Label occluded objects as if they are fully visible.** We see better detections when we label occluded objects as if they are fully visible. This will help the model to learn that the car is still there, even though it is not fully visible.	
3	A vehicle is considered within the ROI if **the center point of its bounding box lies inside the ROI**.	
4	Objects being transported by trucks must be **labeled separately**.	
5	Traffic signs covering vehicles must be **labeled as others**.	
6	Objects moving by pedestrian must be **labeled as others**.	

Questions

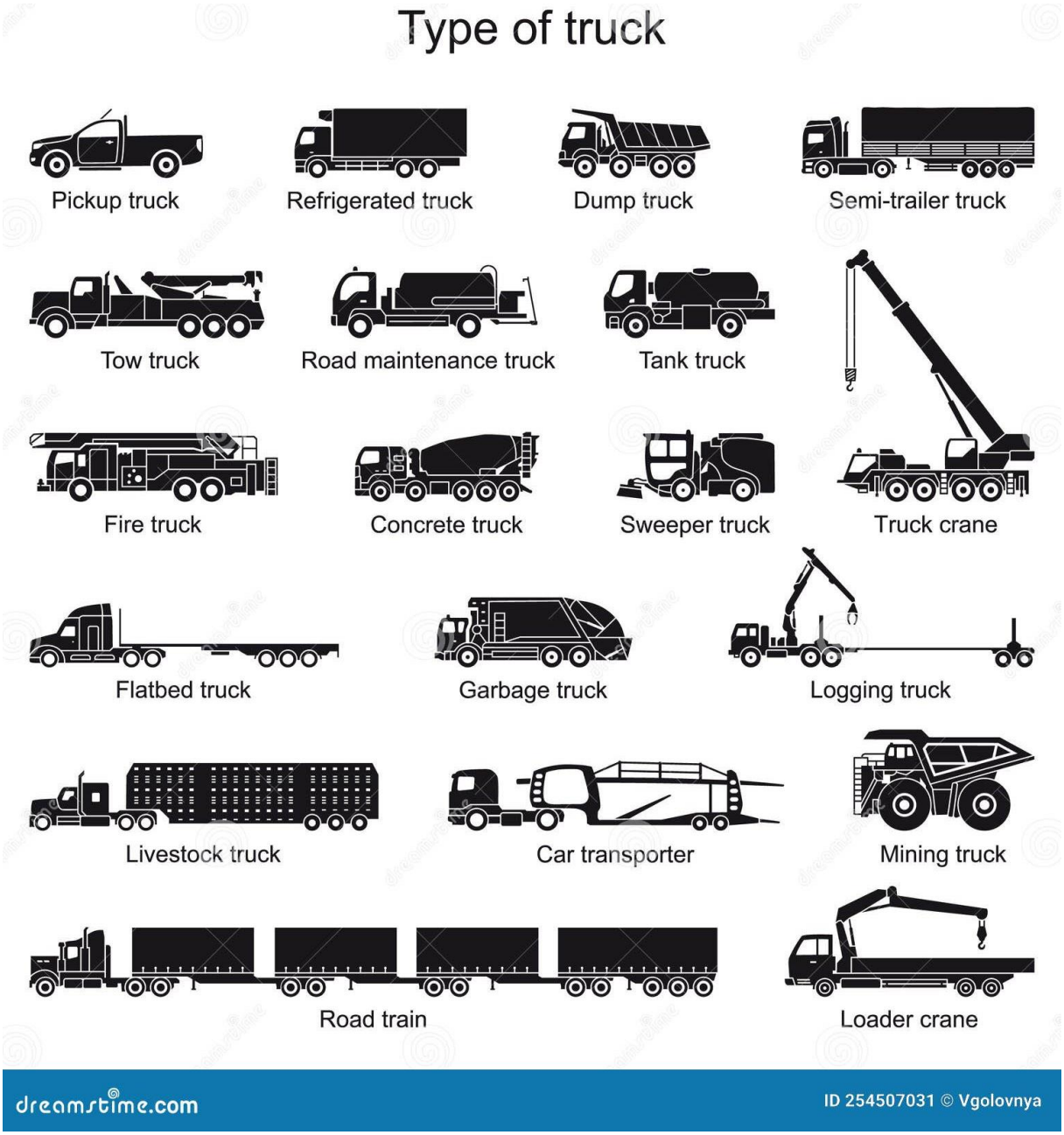
1. **What is the difference between a “pickup truck” and a “small truck”?**



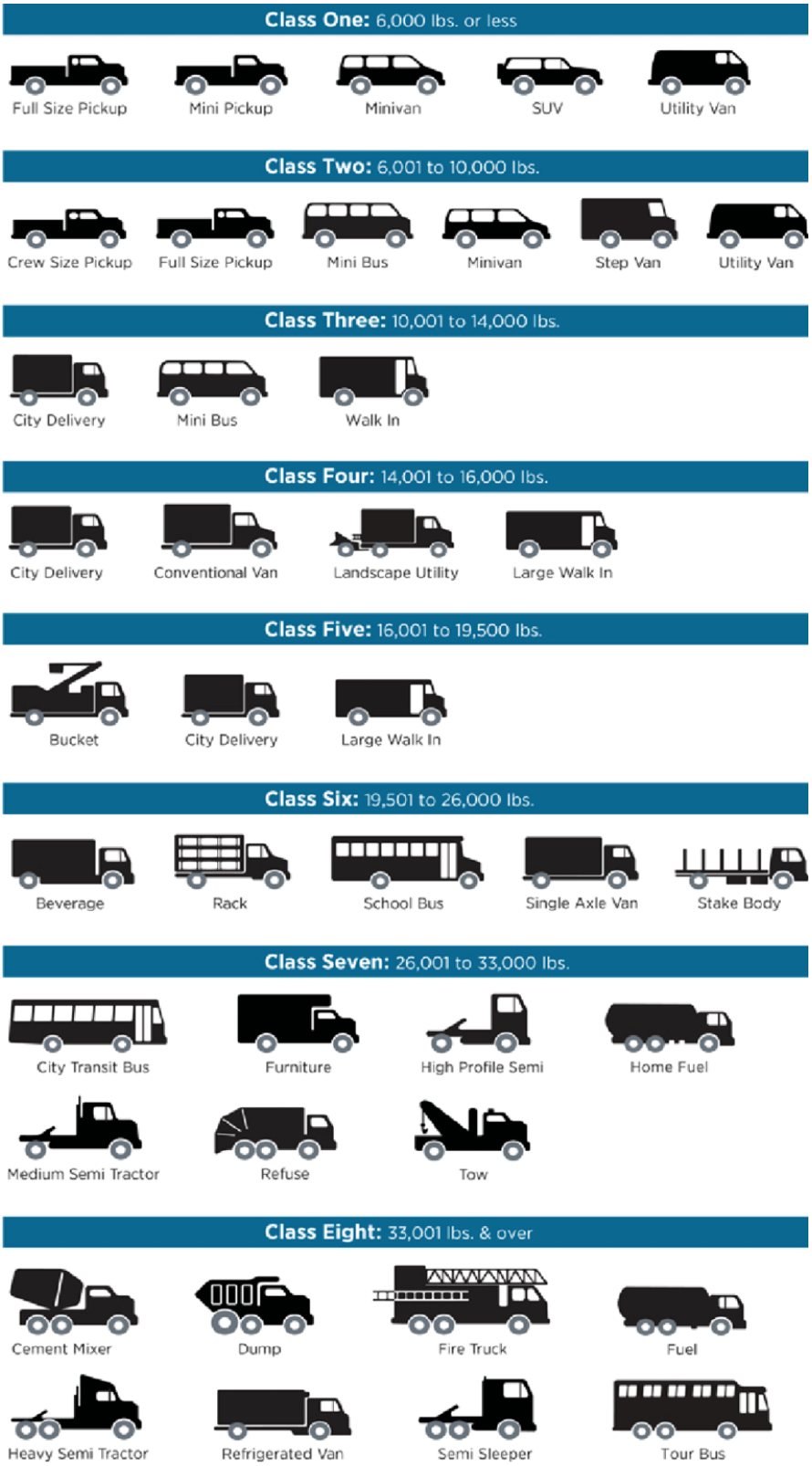
2. **Should we also label objects outside of ROI?**
   - Don’t label → modify training pipeline: https://github.com/ultralytics/ultralytics/issues/16440

References

1. **Types of trucks:**



2. **Vehicle classification based on weights and sizes:**



3. **Links:**
   - https://www.esri.com/arcgis-blog/products/arcgis-pro/geoai/tips-for-labeling-images-for-object-detection-models/
   - https://github.com/waymo-research/waymo-open-dataset/blob/master/docs/labeling_specifications.md